INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Counting Review

Name: _____

The Fundamental Principle of Counting: If a first task can be completed in m ways, and for each way of completing the first task a second task can be completed in n ways, then the total number of ways to complete both tasks in order is $m \cdot n$.

More generally, if k tasks are completed in order, and the tasks can be completed in m_1, m_2, m_3, ..., m_k ways, respectively, no matter how the earlier tasks are completed, then the total number of ways to complete all k tasks in order is

$$m_1 \cdot m_2 \cdot m_3 \cdot \ldots \cdot m_k.$$

Factorials: $n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 2 \cdot 1$, and $0! = 1$. The number of ways of permuting the n elements of an n-element set is $n!$.

Permutations and combinations: The number of ways of permuting k elements of an n-element set, when order matters, is

$$_nP_k = \frac{n!}{(n-k)!}.$$

The number of ways of choosing k elements of an n-element set, when order does not matter, is

$$_nC_k = \frac{n!}{k! \cdot (n-k)!}.$$

1. Seven patients wait in line for flu shots: five adults, Ana, Ben, Carla, David, and Eva, and two children, Kim and Leo. Let U be the set of orders in which the seven patients can line up. Using first initials, one element of U is BKACDLE.

(a) Find $|U|$.

(b) Let F be the set of orders in which the two children are at the front of the line. Find $|F|$.

(c) Let G be the set of orders that begin and end with an adult. Find $|G|$.

2. A nursing student has 3 different anatomy books, A_1, A_2, and A_3, and 4 different chemistry books, C_1, C_2, C_3, and C_4, to put on a shelf in a row. Let U be the set of orders in which she can arrange all 7 books. One element of U is $\mathrm{A}_2\,\mathrm{C}_1\,\mathrm{C}_3\,\mathrm{A}_1\,\mathrm{C}_4\,\mathrm{A}_3\,\mathrm{C}_2$.

(a) Find $|U|$.

(b) Let L be the set of orders in which the anatomy books are all on the left and the chemistry books are all on the right. Find $|L|$.

(c) Let T be the set of orders in which the books on each subject are kept together, with either subject first. Find $|T|$.

(d) Is $L \subseteq T$? Explain.

Counting with Complements: If A is a subset of a universal set U, then

$$|A| = |U| - |A^c|.$$

3. Seven patients wait in line for flu shots: five adults, Ana, Ben, Carla, David, and Eva, and two children, Kim and Leo. Let U be the set of orders in which the seven patients can line up, and let N be the set of orders in which the two children are next to each other.

(a) Find $|N|$.

(b) Describe N^c in words, and find $|N^c|$.

4. A committee of 4 is chosen from 5 nurses, 4 doctors, and 3 pharmacists, and the committee members have no distinct roles. Let U be the set of all possible committees. For example, one element of U is a committee made up of 2 nurses, 1 doctor, and 1 pharmacist. Let M be the set of committees that include members of at least two different professions.

(a) Find $|U|$.

(b) Let N, D, and P be the sets of committees made up entirely of nurses, entirely of doctors, and entirely of pharmacists. Find $|N|$, $|D|$, and $|P|$.

(c) Explain why $M^{\mathrm{c}} = N \cup D \cup P$, and why N, D, and P are disjoint. Find $|M^{\mathrm{c}}|$.

(d) Find $|M|$.

5. A hospital labels its medicine cabinets with 4-letter codes, using the letters A through Z with no letter repeated. The vowels are A, E, I, O, and U. Let U be the set of all possible codes. One element of U is KEYS.

(a) Find $|U|$.

(b) Let V be the set of codes that start and end with a vowel. Find $|V|$.

(c) Let W be the set of codes that contain at least one vowel. Describe W^{c} in words, and find $|W^{\mathrm{c}}|$.

(d) Find $|W|$.

6. A hospital unit has 9 nurses, including Nurse Ruiz. A planning committee needs a head nurse, an assistant head nurse, and 4 other nurses with no distinct roles. No nurse can fill more than one position. Let U be the set of all possible committees, and let R be the set of committees that include Nurse Ruiz.

(a) Find $|U|$.

(b) Describe R^c in words, and find $|R^c|$.

(c) Find $|R|$.

The Inclusion-Exclusion Principle: If A and B are sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

7. The letters of the word NURSE are arranged in a row, using each letter once. The vowels are U and E. Let U be the set of all arrangements; one element of U is SNURE. Let F be the set of arrangements whose first letter is a vowel, and let L be the set of arrangements whose last letter is a vowel.

(a) Find $|U|$.

(b) Find $|F|$, $|L|$, and $|F \cap L|$.

(c) Describe $F \cup L$ in words, and find $|F \cup L|$.

(d) Describe $(F \cup L)^c$ in words, and use it to check your answer to part (c).

The Addition Principle: If A_1, A_2, ..., A_k are disjoint sets, then

$$|A_1 \cup A_2 \cup \ldots \cup A_k| = |A_1| + |A_2| + \ldots + |A_k|.$$

8. A committee of 5 will be chosen from 4 pharmacists and 6 pharmacy technicians, and the committee members have no distinct roles. Let U be the set of all possible committees, and let D be the set of committees with at least 2 pharmacists and at least 2 pharmacy technicians.

(a) For each k, let D_k be the set of committees in D with exactly k pharmacists. Which of the sets D_0, D_1, D_2, D_3, D_4, and D_5 are not empty? Explain.

(b) Find the cardinality of each set from part (a) that is not empty.

(c) Find $|D|$.

9. A pharmacy technician chooses 4 of 10 bottles to inspect, and the order does not matter. Three of the 10 bottles come from a recalled lot. Let U be the set of all possible choices of 4 bottles, and for each k, let E_k be the set of choices that include exactly k recalled bottles.

(a) Find $|U|$.

(b) Find $|E_2|$ and $|E_3|$.

(c) How many choices include at least 2 recalled bottles?

(d) Explain why $(E_2 \cup E_3)^{\mathrm{c}} = E_0 \cup E_1$, and use this to find the answer to part (c) again.

10. A red die, a blue die, and a green die, each with 6 sides numbered 1 through 6, are rolled. A result is recorded as the numbers on the red, blue, and green dice, in that order. Let U be the set of all results, so $(2, 5, 3)$ and $(3, 5, 2)$ are different elements of U. For each whole number s, let S_s be the set of results whose three numbers add up to s.

(a) Find $|U|$.

(b) List every way to write 10 as a sum of three numbers from 1 to 6. Write the three numbers in each sum from smallest to largest, as in $1 + 3 + 6$.

(c) Each sum in part (b) gives a subset of S_{10}: the results that show exactly those three numbers. Find the cardinality of each of these subsets.

(d) Find $|S_{10}|$.

(e) Find $|S_9|$.

11. You can climb a staircase of ten steps by going either one or two steps each time. Let U be the set of ways to climb the staircase. For example, going 2, 1, 1, 2, 2, 1, 1 steps is one element of U. For each k, let T_k be the set of ways to climb the staircase with exactly k two-step moves.

(a) Suppose you take exactly 3 two-step moves. How many one-step moves do you take, and how many moves in all? Find $|T_3|$.

(b) What are the possible numbers of two-step moves?

(c) Find $|T_k|$ for each possible k.

(d) Find $|U|$.

12. A high school dance committee consisting of exactly 14 students is to be chosen from a group of 5 freshmen, 7 sophomores, 10 juniors, and 8 seniors. Let U denote the set of possible committees.

(a) Find $|U|$.

(b) Let A be the set of committees consisting of 2 freshmen, 3 sophomores, 4 juniors, and 5 seniors. Find $|A|$.

(c) Let B be the set of committees with no freshmen. Find $|B|$. How many committees include at least one freshman?

(d) Let C be the set of committees with exactly 2 freshmen. Find $|C|$.

13. 10 students, including Jane, John, Eva, and Ed, are to be arranged in a line for a photo. Let U be the set of possible arrangements.

(a) Find $|U|$.

(b) Let A be the set of arrangements in which Jane and John stand next to each other and Eva and Ed stand next to each other. Find $|A|$.

(c) Let J be the set of arrangements in which Jane and John stand next to each other. Find $|J|$, and explain why $A \subseteq J$.

(d) Let B be the set of arrangements in which Jane and John stand next to each other but Eva and Ed refuse to stand next to each other. Find $|B|$.

14. There are 10 people, A, B, C, D, E, F, G, H, I, and J, on a fundraising committee. The committee needs to choose a chairperson, a secretary, and a treasurer. Let U be the set of possible choices for the three positions.

(a) Find $|U|$.

(b) Let L be the set of choices in which C is the chairperson. Find $|L|$.

(c) Let M be the set of choices in which B is not the chairperson. Find $|M|$.

(d) Let N be the set of choices in which A is not the chairperson or the secretary. Find $|N|$.

(e) Let P be the set of choices in which either I or J is the treasurer. Find $|P|$.

(f) Let Q be the set of choices in which at least one of E, F, and G holds one of the three positions. Find $|Q|$.

15. Let U be the set of strings of 6 letters from the standard 26-letter alphabet, where repetitions are not allowed. For example, QBXAMF is in U, but QBXAMQ is not.

(a) How many strings are there in all?

(b) Let X be the set of strings that contain the letters A and B. Find $|X|$.

(c) Let Y be the set of strings in X in which A and B are consecutive, with A immediately preceding B. Find $|Y|$.

(d) Let Z be the set of strings in X in which A appears to the left of B. Find $|Z|$.

(e) Explain why $Y \subseteq Z \subseteq X$.

16. A clinic has 5 nurses and 4 medical assistants. For a vaccination drive, it needs a team leader, who must be one of the nurses, and 3 other team members with no distinct roles, chosen from the remaining 8 staff members. Let U be the set of all possible teams, and for each k, let M_k be the set of teams with exactly k medical assistants.

(a) Find $|U|$.

(b) How many teams include at least one medical assistant?

(c) A student finds the number of teams with at least one medical assistant this way: "There are 5 ways to choose the leader and 4 ways to choose one medical assistant to be on the team. The other 2 members can be any 2 of the remaining 7 staff members, in ${}_7C_2 = 21$ ways. So there are $5 \cdot 4 \cdot 21 = 420$ teams." Explain what is wrong with this argument.

(d) How many teams include at least 2 medical assistants?